

Memory Test Strategy (iOS NE ceiling + leak soak)

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Master technical specification — Volume 8 (Testing & QA), nano-detail document **memory**, one of the seven §11.4.169 cross-cutting test-type deep-dives. It deepens ~~10-testing-acceptance-and-qa.md~~ §5.12 (MEM) into an implementation-ready bank for the **make-or-break** iOS NEPacketTunnelProvider memory ceiling (Phase-0 gate G3), 24 h leak soaks, and the coordinator's bounded memory at 10k streams (SLO4). SPEC-ONLY: it describes harness, fixtures, evidence, gate, and the paired §1.1 mutation; it does not build the product. Sources cited inline by id — [OVERVIEW] = doc 10; [SYNTHESIS] = v09-research/_SYNTHESIS.md; [04_P0] = Phase-0 spike; [reconcile] = [../v03-control-plane/reconciliation-flow.md]; [TM] = [../v05-security/threat-model.md]. Claims not grounded in the evidence base are marked **UNVERIFIED** per constitution §11.4.6 — never fabricated.

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0. Scope on HelixVPN surfaces — why memory is make-or-break

The single decision the whole client architecture rests on is the **iOS NEPacketTunnelProvider memory ceiling** (~15 MB historical [SYNTHESIS §5]): iOS jetsam-kills a packet-tunnel extension that exceeds its footprint, so the entire choice to write helix-core in **Rust, not Go** (decision D2 [SYNTHESIS §3]) is justified *only* if the Rust core + WG + transport stack stays under that ceiling with $\geq 30\%$ headroom on a real device. The Phase-0 gate **G3** is the make-or-break test for the program: if Rust cannot meet it, the architecture changes [04_P0]. This bank also covers leak-freedom over a 24 h soak (the reconcile loop and the QUIC/MASQUE buffer pool must not grow), Android VpnService RSS, and the coordinator's bounded memory at 10k streams (SLO4).

Surface	Memory claim	Why it matters	Gate
iOS NEPacketTunnelProvider	peak $\text{phys_footprint} < \text{ceiling}$, $\geq 30\%$ headroom	jetsam kills the tunnel above the ceiling	G3 (make-or-break)
helix-core 24 h soak	no RSS growth over sustained transfer	a leak in reconcile/QUIC buffers eventually OOMs the extension	MEM soak
Android VpnService	RSS within the Android background budget	a too-heavy service is killed under memory pressure	MEM
coordinator @ 10k streams	bounded RSS over a 24 h soak	an unbounded per-stream allocation OOMs the gateway	SLO4

1. The four memory claims

(A) G3 — iOS NE peak footprint (make-or-break). On a **real iOS device** (the simulator does not reproduce the jetsam ceiling, decision QA-D3 — overview §10), the Rust core under sustained real traffic must keep `task_vm_info.phys_footprint` (the metric iOS jetsam actually evaluates, **not** resident-size) below the ceiling with $\geq 30\%$ **headroom** ($\text{peak} < \text{ceiling} \times 0.7$). This is the program's go/no-go.

(B) Leak-freedom over a 24 h soak. A sustained real transfer for 24 h must show **no monotonic RSS growth** — a leak in the reconcile loop (each map apply must free the old peer set) or the QUIC/MASQUE buffer pool (buffers must be returned, not accumulated) would be a slow OOM. The claim is a flat (within jitter) RSS-vs-time series, asserted by a monotonic-growth detector with a slope threshold.

(C) **Android VpnService RSS**. The Android tunnel service RSS (via `dumpsys meminfo <pkg>`) stays within the platform background budget so Android does not kill it under memory pressure.

(D) **SLO4 — coordinator bounded memory at 10k streams**. $N=10k$ simulated agents holding `WatchNetworkMap` streams must keep coordinator RSS **bounded** over a 24 h soak — per-stream state is $O(1)$ and the in-memory topology graph is shared, not copied per stream ([reconcile], [TM T-COORD-D-1]). An unbounded per-stream allocation is the OOM vector.

2. Harness — per-platform RSS samplers

Memory is platform-specific (§11.4.81 cross-platform parity); each platform has its OS-correct sampler, chosen at runtime. **No mocks** — the metric must come from the real OS.

Platform	Sampler	Metric (the jetsam/ OOM-relevant one)
iOS NE	<code>task_info(TASK_VM_INFO) → phys_footprint</code> ; Instruments Allocations/Leaks .trace	<code>phys_footprint</code> (the jetsam metric, not resident size)
Linux (helix-core, edge, coordinator)	<code>/proc/<pid>/status VmRSS / smaps_rollopp Pss</code>	RSS / Pss over time
Android VpnService	<code>dumpsys meminfo <pkg></code> (TOTAL PSS)	PSS in the background budget
coordinator soak	<code>/proc/<pid>/status</code> at 1-min cadence over 24 h	RSS bounded / no growth

```
// shims/apple/PacketTunnelProvider+MemoryProbe.swift – G3
    evidence emitter (overview §5.12)
func sampleFootprint() -> UInt64 { // bytes;
    appended to qa-results/mem/ios_rss.csv
    var info = task_vm_info_data_t()
    var count =
        mach_msg_type_number_t(MemoryLayout<task_vm_info>.size /
        MemoryLayout<integer_t>.size)
    let kr = withUnsafeMutablePointer(to: &info) {
        $0.withMemoryRebound(to: integer_t.self, capacity:
        Int(count)) {
            task_info(mach_task_self_,
            task_flavor_t(TASK_VM_INFO), $0, &count) } }
    return kr == KERN_SUCCESS ? info.phys_footprint : 0 //
        phys_footprint == the jetsam metric
}
```

The samplers themselves must be light (Heisenberg constraint, §11.4.24-class): the probe samples at a fixed cadence (≤ 1 Hz) and writes a CSV out-of-band; it must not itself perturb the footprint it measures. The bank runs under `make test MEM` stage for short probes and a backgrounded 24 h soak job (§11.4.89). Coordinator soak boots PG+Redis via containers (§11.4.76, rootless §11.4.161).

3. Fixtures — real device, real soak (§11.4.27)

Fixture	What	Why real
real iOS device (paired)	a physical iPhone running the NE build	the jetsam ceiling does not reproduce in the simulator (QA-D3)
device_build.ipa	the shipping NE extension build (release config)	G3 must measure the <i>shipping</i> artifact (§11.4.108), not a debug build with allocator debug overhead
sustained real transfer	a real iperf3/file transfer through the tunnel	a leak shows only under real buffer churn, not an idle tunnel
10k agent-fuzzer	a real load generator opening 10k WatchNetworkMap streams	SLO4 needs real streams holding real per-stream state

If **no real iOS device is available**, G3 is an honest §11.4.3 SKIP: hardware_not_present with a tracked operator-attended migration item (QA-D3, overview §10) — **never** a simulator PASS faked as a device result (that would be a B3 wrong-plane bluff).

4. Captured evidence (§11.4.69 / .107)

Every PASS cites artifacts under qa-results/mem/<run-id>/. The §11.4.69 class is MEM. §11.4.107 liveness applies: the RSS series must be over a **window** during *real, advancing* traffic (a single footprint sample while the tunnel is idle is not proof — the core must be doing real work).

Test	Artifact	Asserts
G3-IOS-PEAK	ios_rss.csv + Instruments .trace	peak phys_footprint < ceiling × 0.7 (≥30% headroom)
MEM-SOAK-LEAK	core_rss_24h.csv	no monotonic growth (slope < threshold) over 24 h
MEM-ANDROID-RSS	android_meminfo.csv	PSS within the background budget
SLO4-COORD-BOUNDED	coord_rss_24h.csv + stream_count.csv	RSS bounded while holding 10k streams over 24 h

The evidence is the **time series**, not a point: a flat-or-bounded RSS-vs-time curve under load. A single in-bounds sample is a B-class point-not-window bluff (§11.4.107(1)); the analyzer asserts the *peak over the window* and the *growth slope*, both self-validated (§5, golden-good flat series PASSes, golden-bad leaking series FAILs).

5. Determinism (§11.4.50)

The G3 peak and the soak slope are asserted N=3 (G3 short probe) / over a single authoritative 24 h run (soak — a 24 h run is not repeated N=3, but its *verdict* is deterministic: peak < ceiling is a fact about that run, and the run is re-run on any regression). `ab_run_n_times "g3-ios-peak" 3 run_ios_probe` runs the short G3 probe 3× against the same .ipa MD5 + same device; the evidence-hash is over (peak_footprint_bucket, headroom_pct_meets_30: bool) — all 3 MUST agree. A run where the headroom meets 30% in 2 of 3 is **auto-FAIL** (the ceiling margin must be reliable, not lucky). The soak's monotonic-growth verdict is deterministic given the same workload; a divergent slope across two soaks is a regression triggering §11.4.4 STOP.

6. Acceptance gate

Gate	Bar	Evidence	Phase
G3 (make-or-break)	peak phys_footprint < ceiling, ≥30% headroom, on a real device	ios_rss.csv + .trace	Phase 0 (release-blocking go/no-go)
MEM-SOAK-LEAK	no monotonic RSS growth over 24 h under load	core_rss_24h.csv	MVP
MEM-ANDROID-RSS	PSS within background budget	android_meminfo.csv	MVP
SLO4	coordinator RSS bounded @ 10k streams / 24 h	coord_rss_24h.csv	MVP (release-blocking SLO)

G3 is the program's make-or-break gate: a No-Go means the architecture changes (Rust→? or a memory budget rework), escalated per §11.4.66 — never silently overrun [04_P0]. G3 + SLO4 are §11.4.132 risk-ordered high (G3 is the highest-risk Phase-0 gate). The MEM cell appears in the ledger for F-IOS-NE-MEM (G3) and SLO4 (overview §6.3, §7.2).

7. The paired §1.1 mutation

MUTATION (paired §1.1, gate CM-MEM-LEAK-DETECTOR-SELFVALIDATED):

Replace the golden-bad leaking RSS series fixture with a flat one (or

loosen the monotonic-growth slope threshold to accept any slope).

EXPECTED: the leak-detector self-validation test that asserts the golden-bad (leaking) series FAILs now PASSES the leak → meta-test FAILs → mutation caught (defeats B5 – an analyzer

that passes its golden-bad fixture is the bluff).

RESTORE: re-instate the leaking golden-bad fixture / strict slope; re-run → GREEN.

A second mutation (CM-SLO4-BOUNDED) introduces a per-stream allocation that is never freed in the coordinator fan-out; expected: coord_rss_24h.csv shows unbounded growth at 10k streams → SLO4 FAILs → caught (this is the real OOM vector the gate guards). A third (CM-G3-METRIC-IS-FOOTPRINT) swaps the G3 metric from phys_footprint to resident-size (which under-reports the jetsam-relevant footprint); expected: the metric-correctness check FAILs because the asserted metric no longer matches the jetsam metric (defeats a B3 wrong-plane bluff). All mutations restored, tree verified quiescent (§11.4.84) before commit.

8. Test skeletons

```
# rig/ios_g3_probe.sh – G3-IOS-PEAK on a REAL device (QA-D3),
# §11.4.108 shipping artifact
set -euo pipefail
out="qa-results/mem/g3_$(date +%s)"; mkdir -p "$out"
device_serial=$(ios_paired_device) || ab_skip_with_reason "G3 iOS
peak" hardware_not_present # §11.4.3, never a sim PASS
ios_install "$device_serial" device_build.ipa
# the RELEASE NE build
ios_start_tunnel "$device_serial"
ios_run_sustained_transfer "$device_serial" 600 &
# 10 min of REAL traffic (liveness §11.4.107)
ios_sample_footprint_1hz "$device_serial" "$out/ios_rss.csv"
# phys_footprint series
ios_capture_instruments_trace "$device_serial" "$out/leaks.trace"
peak=$(max_col "$out/ios_rss.csv" phys_footprint)
ceiling=$NE_CEILING_BYTES
awk -v p="$peak" -v c="$ceiling" 'BEGIN{exit !(p < c*0.7)}' \
&& ab_pass_with_evidence "G3: peak $peak < ceiling $ceiling
(>=30% headroom)" "$out" \
|| ab_fail "G3 NO-GO: peak $peak exceeds 70% of ceiling
$ceiling – escalate (§11.4.66)"

# scripts/mem_soak.sh – MEM-SOAK-LEAK 24 h, no monotonic growth
# (backgrounded §11.4.89)
set -euo pipefail
out="qa-results/mem/soak_$(date +%s)"; mkdir -p "$out"
```

```

trap 'stop_transfer; rig/netns_down.sh' EXIT
start_sustained_transfer &
sample_rss_1min "$(core_pid)" "$out/core_rss_24h.csv" 86400
    # 24 h at 1-min cadence
slope=$(monotonic_growth_slope "$out/core_rss_24h.csv")
    # bytes/hour
within_threshold "$slope" \
    && ab_pass_with_evidence "no RSS leak over 24h (slope=$slope B/h)" "$out" \
    || ab_fail "RSS leak: slope=$slope B/h exceeds threshold –
    §11.4.4 STOP + systematic-debug"

// helix-go/internal/coordinator/slo4_soak_test.go – SL04 bounded
// @ 10k streams
func TestCoordinatorMemoryBoundedAt10kStreams(t *testing.T) {
    infra := mustBootRedisPG(t) //
    containers, rootless §11.4.161
    coord := startCoordinator(t, infra)
    agents := openNStreams(t, coord, 10_000) // 10k
    real WatchNetworkMap streams
    rss := sampleRSSOver(t, coord.Pid(), 24*time.Hour,
        time.Minute)
    requireBounded(t, rss, slo4Ceiling) // RSS
    bounded, no unbounded per-stream growth
    writeEvidence(t, "qa-results/mem/coord_rss_24h.csv", rss)
    _ = agents
    // paired §1.1 (CM-SLO4-BOUNDED): leak a per-stream allocation
    → unbounded growth → SLO4 FAILs
}

```

Honest boundary (§11.4.6). The exact NE ceiling value (NE_CEILING_BYTES) is a platform constant cited as ~15 MB historical [SYNTHESIS §5] but is **UNVERIFIED** for the current iOS version on the target device class until the G3 probe runs on the real device — the ceiling is iOS-version and device-dependent, and the spec uses the conservative historical figure. The 24 h soak proves no leak *over that window under that workload*; a leak with a period longer than 24 h, or under a workload not exercised, is an §11.4.118 discovery gap, not a proven absence. SLO4's bound at exactly 10k streams is the MVP target; the convergence-and-memory behaviour beyond 10k is a Phase-2 SCALE concern ([TM T-COORD-D-1] notes the 10k-stream p99 is itself **UNVERIFIED** until the soak runs).

Sources verified

- [OVERVIEW] [../10-testing-acceptance-and-qa.md] — §5.12 (MEM strategy + sampleFootprint skeleton + real-device requirement), §6.3 (F-IOS-NE-MEM ledger), §7.1 (G3 gate), §7.2 (SLO4), §8 (determinism/risk-order), §10 QA-D3. (Read 2026-06-26.)
- [SYNTHESIS] v09-research/_SYNTHESIS.md — §3 (D2 Rust-not-Go decision), §5 (~15 MB NE ceiling). (Cited via overview.)
- [04_P0] 04_VPN_CLD/HelixVPN-Phase0-Spike.md — G3 make-or-break iOS memory gate, ≥30% headroom. (Cited via overview.)

- [reconcile] [../v03-control-plane/reconciliation-flow.md] — coordinator topology graph + stream fan-out (SLO4 per-stream state). (Read 2026-06-26.)
- [TM] [../v05-security/threat-model.md] — T-COORD-D-1 (10k-stream p99/memory **UNVERIFIED** soak). (Read 2026-06-26.)
- Constitution: §11.4.169, §11.4.27 (real device/no mocks), §11.4.3 (honest SKIP hardware_not_present), §11.4.69 (MEM evidence class), §11.4.107(1)/(10) (window-not-point liveness + self-validated analyzer), §11.4.50 (determinism), §11.4.81 (per-platform sampler), §11.4.108 (shipping artifact), §11.4.132 (G3/SLO4 high-risk), §11.4.118 (discovery boundary), §11.4.84 (quiescence), §11.4.66 (escalate G3 No-Go), §1.1 (paired mutation).